

Corrigendum to “Extending Flexible Boolean Semantics for the Language of Mathematics”

Shashank Pathak^[0000–0003–4519–3689]

Department of Computer Science, The University of Manchester, UK
`shashank.pathak@manchester.ac.uk`

This note clarifies one aspect of the analysis presented in [Pat26]. In that paper (p. 161), I stated that Winter’s [Win02] framework generates three readings for the sentence:

- (1) The integers and the reals are countable.

The three readings are the following.

- (2) $((\mathbf{E}[\![\text{the integers}]\!]) \sqcap (\mathbf{E}[\![\text{the reals}]\!]))(\mathbf{countable})$.
- (3) $\mathbf{E}([\![\text{the integers}]\!] \sqcap [\![\text{the reals}]\!])(\mathbf{countable})$.
- (4) $\mathbf{min}((\mathbf{E}[\![\text{the integers}]\!]) \sqcap (\mathbf{E}[\![\text{the reals}]\!]))(\mathbf{countable})$.

A closer reanalysis shows that only two readings can be derived. The third reading cannot be derived as **min** cannot be used after **E** in the subject position [Sch01, p. 4]. Accordingly, there are the following two possible readings.

- (5) a. $((\mathbf{E}[\![\text{the integers}]\!]) \sqcap (\mathbf{E}[\![\text{the reals}]\!]))(\mathbf{countable})$.
- b. $\mathbf{E}([\![\text{the integers}]\!] \sqcap [\![\text{the reals}]\!])(\mathbf{countable})$.

The fact that there are two readings rather than three does not affect the central theoretical claim of the paper. The main argument was that the framework overgenerates interpretations, and this issue remains since the two readings in question are still derivable. The overgeneration issue remains because the determiner phrase *the integers and the reals* is syntactically ambiguous based on the toy grammar given in [Win02, p. 186]. The ambiguity arises because the coordinator *and* can coordinate either determiner phrases or noun phrases, which gives rise to the two distinct readings. The corresponding syntactic structures are shown in Figures 1 and 2, respectively.

The purpose of this note has been to provide a more accurate statement of the earlier claim. I regret the inaccuracy in the original formulation and hope this clarification helps prevent any misunderstanding.

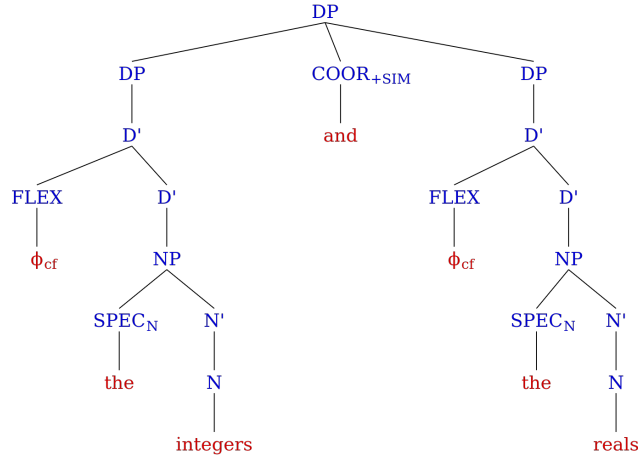


Fig. 1. Syntactic Structure of the DP *The integers and the reals* resulting in reading (2).

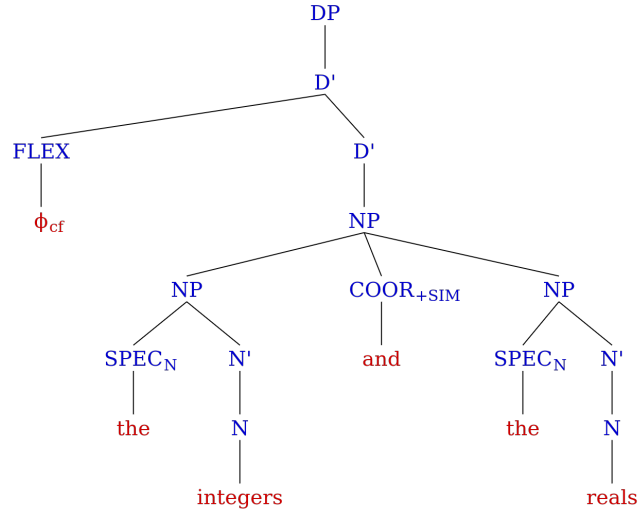


Fig. 2. Syntactic Structure of the DP *The integers and the reals* resulting in reading (3).

References

- Pat26. Shashank Pathak. Extending flexible boolean semantics for the language of mathematics. In Valeria de Paiva and Peter Koepke, editors, *Intelligent Computer Mathematics*, pages 152–170, Cham, 2026. Springer Nature Switzerland.

- Sch01. Roger Schwarzschild. Review of winter (1998) flexible boolean semantics: Coordination, plurality and scope in natural language. *Glott International* 5.4, 2001.
- Win02. Yoad Winter. *Flexibility Principles in Boolean Semantics: The Interpretation of Coordination, Plurality, and Scope in Natural Language*. The MIT Press, 02 2002.